

Improving California's Freeway Contractor Evaluation Framework

Group #1

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Caltrans primarily uses lowest-bid procurement for highway construction and maintenance contracts. While cost efficiency is important, this approach can contribute to schedule delays, quality deficiencies, costly rework, and increased financial risk to California taxpayers. Group #1 conducted a 10-week, five-member consulting project to develop the California Freeway Contractor Evaluation Framework (CFCEF), a performance-based contractor evaluation model that considers price alongside schedule, cost control, project management, quality, financial stability, and safety. Key stakeholders include Caltrans, contractors, taxpayers, and communities affected by highway construction.

Objectives

The project sought to evaluate Caltrans' current lowest-bid approach, examine research supporting best-value procurement, identify applicable statutory and regulatory requirements, develop a standardized weighted contractor evaluation methodology, and provide Caltrans with a policy recommendation supported by project documentation, a final report, and presentation.

Methods

The project was limited to research, framework design, and policy recommendation and excluded implementation, pilot testing, legislative drafting, contractor interviews, and proprietary data collection. During Planning, the team established a Work Breakdown Structure, 10-week schedule, four milestones, 520 labor hours, and a **\$29,700 budget**, supported by a five-member RACI-based team structure.

During Execution, Microsoft Teams, email, and SharePoint supported communication and document control. The team also implemented stakeholder engagement and escalation

procedures. Monitoring and Control included Earned Value Management, risk and issue tracking, milestone-based quality reviews, and 14 performance metrics covering schedule, cost, scope, quality, stakeholder engagement, and risk.

Outcomes

The project delivered seven approved outputs: a current-state assessment, literature review, Caltrans procurement analysis, CFCEF framework, policy recommendation, project management documentation, and final report and presentation. Collectively, these deliverables provide an evidence-based alternative to price-only contractor selection while maintaining defined controls for scope, schedule, cost, quality, and stakeholder requirements.

Significance

The CFCEF provides Caltrans with a structured and defensible approach to evaluating contractor qualifications and past performance in addition to price. If adopted, the framework could improve contractor accountability, reduce rework and project delays, minimize construction-related disruptions, and strengthen public confidence in the management of California's transportation infrastructure investments. Transportation agencies benefit from a more defensible, evidence-based selection process, while qualified contractors benefit from fairer recognition of reliable performance rather than losing work on price alone.

Initiation Phase

1.1 Define Project Concept (Lewis & Jocelyne)

California's interstate and highway infrastructure relies on a contract bidding process that awards work largely to the lowest-cost bidder rather than the contractor best equipped to deliver quality results on time. This approach has produced a persistent pattern of missed deadlines, substandard workmanship, and projects that must be re-funded and redone costing the state millions of dollars beyond its already substantial annual investment in highway construction and maintenance. Because contracts are frequently paid out before work is verified, financial risk falls disproportionately on California taxpayers rather than on the contractors responsible for the deficiencies.

This project proposes a standardized contractor evaluation process for Caltrans that incorporates multiple performance metrics including schedule adherence, cost control, project management, quality, financial stability, and safety, rather than relying on price alone. The opportunity is significant: a stronger, performance-based bidding model could reduce duplicate spending on redone work, shorten the traffic congestion and community disruption caused by prolonged or repeated roadwork, and shift accountability back onto contractors. Key stakeholders include Caltrans as both sponsor and awarding agency, contractors and construction firms bidding on projects, California taxpayers funding the work, and the commuters and communities affected by construction delays.

A peer reviewed article titled Best-value procurement in design-bid-build construction projects: Empirical analysis of selection outcomes (Gransberg, D. D., & Ellicott, M. A., 2018)

analyzed 162 public institutional best-value procured design-bid-build projects, including evaluation results of 684 contractor proposals, and found that best-value-selected contractors offer substantial qualifications benefits compared with the lowest and average bidders. That directly supports this project's core claim that evaluating contractors on more than price leads to better-qualified selections, which is the empirical backbone of the scope.

For transportation agencies, the expected benefit is a more defensible and predictable selection process that reduces rework, shortens project timelines, and protects public funds from being spent twice on the same work. For contractors, a performance-based model rewards demonstrated qualifications and past performance rather than the lowest price alone, giving well-qualified firms a fairer opportunity to compete and be recognized for reliable, high-quality work.

1.2 Identify Initial Stakeholders and Sponsor

There are several stakeholders and sponsors for this project. Stakeholders consist of Caltrans (California Department of Transportation). This is the agency responsible for awarding contracts, setting procurement standards, and managing statewide highway infrastructure. Then come contractors and construction firms.

The bidders are directly affected by changes to the evaluation and selection criteria. California taxpayers will be the funding source for these projects, bearing the financial risk when contracts are refunded due to poor performance. Commuters and local communities are directly impacted by construction delays, prolonged traffic congestion, and repeated roadwork.

Sponsors include Caltrans. As the state agency with the authority and budgetary control to revise and implement a new contractor evaluation and bidding process for California's interstate and highway infrastructure projects.

1.3 Alignment with Organizational or Societal Needs

Our project addresses the contract bidding process for California's interstate and highway infrastructure, and the persistent problems it creates, including contractors who miss deadlines, deliver low-quality work, or both. This is a longstanding issue rooted in how contracts are awarded largely to the lowest-cost bidder rather than the bidder best equipped to deliver quality results on time. That approach raises serious concerns about workmanship, timeliness, and even the materials used on state infrastructure. California already invests billions of dollars in highway construction and maintenance yet continues to spend millions more re-funding projects that were done poorly or left incomplete the first time.

This project addresses several interconnected societal challenges:

Budget efficiency. California allocates millions of dollars in grants statewide for highway maintenance and development. When contracts are awarded without sufficient accountability for quality, that money is frequently spent twice once on the original (deficient) work, and again to fix or redo it. A bidding process with stronger quality standards built in would reduce how often projects need to be revisited, freeing up funds for other pressing needs across the state.

Traffic congestion and community impact. Construction and repair work on freeways and interstates causes significant traffic delays. When projects run over deadlines or need to be redone due to poor quality, congestion is prolonged unnecessarily. This has real effects on communities' longer commutes, disrupted daily routines, and more time lost to travel that could otherwise go toward work, family, or rest.

Financial risk and accountability. Contracts are often paid out, in whole or in part, before work is completed or verified. If the finished work does not meet quality standards, that money is effectively lost, and the state must allocate additional funds to have the job redone. This "pay first, verify later" structure shifts financial risk onto taxpayers rather than the contractors responsible for the work, and it compounds the budget problem described above dollars that could go toward other state priorities instead go toward paying for the same job twice.

Taken together, these challenges show that the current bidding model does not just create isolated project delays it has a compounding effect on the state budget, on public infrastructure reliability, and on the daily lives of Californians who depend on these roads.

1.4 Rationale for Project Selection (Donavin)

Our group decided to complete this assignment because California's transportation system is vital to the movement of people, goods, and services throughout the state. California's freeway system is some of the busiest in the nation and needs continued maintenance, expansion, and modernization. Agencies at the state, county, and city levels use millions of dollars of taxpayer funds each year on transportation improvement projects with the intention of reducing congestion, increasing safety, and improving reliability. Too many freeway projects face schedule delays, cost overruns, and quality issues that can have negative impacts on both public agencies and citizens. Contractor performance is only one of many variables that affect project success, which is why improving the contractor selection process will increase the likelihood that freeway projects are completed on-time and within budget (California Department of Transportation [Caltrans], 2024).

Another reason we selected this project is that we believe it presents a real-world problem that affects millions of Californians every day. Low-bid contracting protects public agencies by encouraging price competition. However, the lowest bid does not prove that a contractor is best-suited to perform the work at the required quality, schedule, safety, and project management levels. California's Freeway Contractor Evaluation Framework (CFCEF) therefore would not discount price from the decision; it would simply weigh price against other demonstrated performance factors like schedule compliance, cost control, quality history, financial stability, safety history, and project management experience. Establishing standardized and measurable criteria would allow Caltrans to more objectively compare bidders and increase transparency and accountability during contractor selection (California Department of Transportation [Caltrans], 2024).

We decided to complete this project because it could incorporate several key project management topics. Marchewka (2015) discusses how successful projects should be chosen based on their ability to create value for an organization and support long-term strategic goals, not just their ability to meet an organization's immediate needs. If this information was applied to California's transportation system, readers could see how contractor selection has the power to influence project success, risk, and an agency's ability to meet long-term objectives. Creating a process that ranks contractors based on quantitative performance metrics will allow agencies to choose contractors with a proven track record of performing well in those categories. This topic can tie into key project management concepts such as procurement, stakeholder, quality, and risk management. Project management professionals implement these best practices to help manage stakeholders' expectations, project risks, project scope, project quality, and project procurement.

Establishing a contractor evaluation process that takes many of these factors into account before work begins will help agencies make informed decisions that lead to project success.

The last reason we chose this topic is that credible information can be easily obtained. Caltrans has up-to-date strategic plans, procurement guidance, construction standards, and performance reports that showcase California's infrastructure priorities and how transportation projects should be delivered (Caltrans, 2024). Counties and cities also post similar information on their websites. Access to this data will allow our group to form an unbiased opinion of how contractors are currently selected by reviewing the processes already in place. The fact that we can acquire most of our information from government websites adds credibility to this project because we can base our recommendations off California's transportation policies and standards.

Completing this assignment will allow our group to learn about many project management techniques that can be applied to a real-world public-sector procurement issue. We hope to create a valuable recommendation that Caltrans can use to improve contractor selection practices by establishing uniform evaluation criteria and a weighted-point system as opposed to deferring to lowest price. The scope of this project will be limited to researching current practices, creating a framework, and providing policy recommendations. Implementation and pilot testing would require approval from Caltrans at a later date. For the purposes of this class, this project allows us to review how procurement, stakeholder, quality, and risk management can all be applied to creating a single decision-making tool that could positively impact project success, taxpayer accountability, and public perception.

Feedback from stakeholders will help shape project decisions as CFCEF progresses from initiation into planning and execution phases. Stakeholders such as Caltrans, contractors, commuters, taxpayers, and impacted communities can offer insight into what contractor-

performance areas should be prioritized, confirm if proposed evaluation criteria are clear and feasible, and provide information not visible through procurement documents alone. The team will record feedback trends and compare them against government guidance and research before amending the framework. Suggestions that fall within the approved scope can be considered for future framework drafts. Proposals that change scope, schedule, cost, or expected project deliverables will be processed through formal change control. This process allows stakeholder engagement to feed into project decisions without a single comment derailing the evidence-based intent of the CFCEF (Project Management Institute [PMI], 2021).

Planning Phase

2.1 Define Scope and Deliverables (Lewis)

Scope

This project is scoped to research, design, and recommend a standardized contractor evaluation framework for Caltrans, the California Freeway Contractor Evaluation Framework (CFCEF), as an alternative to the agency's current lowest-bid contractor selection model. Consistent with Marchewka's (2015) view that project planning establishes a project's objectives, scope, boundaries, and resources before execution; the work is organized into two phases: Planning and Execution. A five-member consulting team will complete the project over a 10-week period.

During the Planning phase, the team will define the project scope, develop project management artifacts (including the Work Breakdown Structure, RACI matrix, schedule, and risk register), assess Caltrans' existing contractor selection process, review academic and industry literature on best-value procurement, and analyze publicly available Caltrans

procurement policies to identify statutory and regulatory considerations. These activities establish the research foundation for the project.

The Execution phase focuses on developing the CFCEF by defining evaluation criteria and a weighted scoring methodology based on contractor performance metrics such as cost, schedule, quality, financial stability, and safety. Using the research completed during the Planning phase, the team will prepare a policy recommendation describing how Caltrans could adopt the framework while considering identified legal and regulatory constraints. The project concludes with the preparation of a comprehensive final report and presentation summarizing the research, framework, and recommendations.

The project is intentionally limited to research, framework design, and policy recommendation development. It does not include implementation or pilot testing of the CFCEF, drafting legislation or formal regulatory filings, conducting contractor interviews or field inspections, or collecting proprietary contractor performance data. All analyses will rely on publicly available government and industry sources. In addition, the project addresses only the contractor selection stage of procurement and does not extend to project design, construction management, contract administration, or post-award performance monitoring. Defining these boundaries ensures the project remains achievable within the approved 10-week schedule and budget while producing a practical, evidence-based recommendation for Caltrans.

Deliverables

The project will produce seven primary deliverables aligned with the Planning and Execution phases:

1. **Current-State Assessment** – An evaluation of Caltrans' existing lowest-bid contractor selection process, documenting its strengths, limitations, and opportunities for improvement.
2. **Literature Review and Industry Guidance Summary** – A synthesis of academic research and government guidance on best-value procurement and contractor performance evaluation, including the findings of Gransberg and Ellicott (2018).
3. **Caltrans Procurement Analysis** – An assessment of the statutory and regulatory requirements influencing the adoption of a performance-based contractor evaluation framework.
4. **California Freeway Contractor Evaluation Framework (CFCEF)** – A standardized contractor evaluation model featuring performance criteria, weighted scoring, and an objective evaluation methodology.
5. **Policy Recommendation Report** – A formal recommendation outlining how Caltrans could adopt the CFCEF, including implementation considerations and identified regulatory constraints.
6. **Project Management Documentation** – Project planning and control artifacts, including the Work Breakdown Structure (WBS), RACI matrix, project schedule, and risk register.
7. **Final Report and Presentation** – A comprehensive report and presentation summarizing the project objectives, research findings, proposed framework, recommendations, and overall project outcomes.

2.2 Work Breakdown Structure (Pujan)

A Work Breakdown Structure decomposes the total scope of project into smaller, more manageable components, allowing the team to plan, estimate, assign, and track work at a level

of detail that a high-level project description cannot provide on its own. As Marchewka (2015) explains, breaking a project down into discrete work packages helps ensure that all requirements are captured, provides a foundation for estimating time and cost, and clarifies which team member is accountable for each piece of work. For the project, the WBS is developed after the scope and deliverables are finalized, ensuring that every work package traces directly back to an agreed upon deliverable rather than to assumed work.

The WBS is organized into two top-level phases, Planning and Execution, consistent with the project's overall approach. Each phase is broken into work packages that correspond directly to the work packages used in the project's time and cost estimates, so that the schedule, budget, and scope documentation all reference the same work.

Planning

The project planning and scope definition confirms the boundaries established in the project scope. This package also produces the RACI matrix that assigns responsibility, accountability, consultation, and information roles across the team, and finalizes any outstanding details carried over from the project charter developed during the initiation phase.

The current states assessment documents how Caltrans currently selects contractors under its lowest-bid model, including the process' strengths and its documented weaknesses. This assessment establishes the baseline that the proposed framework is designed to improve upon.

Literature review and Industry Guidance gathers academic and industry sources on procurement, including Gransberg and Ellicott's (2018) empirical analysis of best-value selection outcomes, along with relevant guidance on contractor evaluation practices.

Caltrans procurement analysis reviews Caltrans' publicly available procurement policies and documentation to identify the statutory and regulatory constraints that would affect adoption

of a performance-based model. This ensures the final recommendation is grounded in what Caltrans could feasibly implement.

Execution

Performance-based framework design uses the findings from the literature review and procurement analysis to define the framework's evaluation criteria and to develop a weighted scoring methodology.

Policy Recommendation Development translates the completed framework into an actionable recommendation for Caltrans, outlining a proposed implementation approach, anticipated benefits, and regulatory considerations identified earlier that Caltrans would need to address.

Documentation and Final Presentation compile the current-state assessment, literature review, and framework design into the final written report, finalizes the supporting project management artifacts, and prepares the final presentation summarizing the project's objectives, research, framework, and recommendations.

Overall, structuring the WBS this way keeps the planning phase focused on building the evidence and boundaries the project needs before any design work begins, and keeps the execution phase focused on producing the framework and recommendation that answer the project's original objective.

2.3 Estimate Time & Costs (Donavin)

Estimating realistic time and cost parameters is an important step in the project planning process. Developing these estimates helps manage stakeholder expectations, aids in resource allocation, and creates a baseline for planning the project schedule and budget. Project planning requires "estimating the effort, duration, and resources required to complete each major project

activity before the work starts” (Marchewka, 2015). The estimates created for the California Freeway Contractor Evaluation Framework (CFCEF) project reflect the effort required to research, analyze Caltrans’ current procurement process, design and validate a standardized contractor evaluation framework, and prepare final documentation. These estimates should not be confused with the cost of building a highway and associated contract billing rates. Instead, they reflect on the anticipated effort it will take a consulting team to produce an evidence-based framework and policy recommendation.

To generate time and cost estimates, the project team used expert judgment and bottom-up estimating. Each work package identified in the work breakdown structure (WBS) was given an individual estimate. Then, the cumulative cost and effort were calculated to reach a project-wide estimate (Marchewka, 2015). Bottom-up estimates help increase accuracy and allow for more granular resource allocation. They can also be used to create a more realistic schedule and cost baseline. The time and cost estimates also support the constraints of project management’s Triple Constraint by allowing project managers to balance scope, schedule, and cost during the Planning Phase of the project.

Labor is the driving cost of this project due to the research-intensive nature of the work. Because most of the information used to construct the CFCEF will be sourced from Caltrans’ publicly available project publications, FHWA guidance, and scholarly articles, person-hours represent the project’s greatest expense. Activities included in the 520 labor hours include completion of the current-state assessment, literature review, procurement process analysis, framework design, policy recommendation, compilation into a final report, and creation of a final presentation. The person-hours will be split among members of the project team, allowing

multiple activities to take place simultaneously while keeping the project on the proposed ten-week timeline.

Table 1

Estimated Project Time

Work Package	Duration	Primary Resource
Project Planning & Scope Definition	1 week	Project Manager
Current-State Assessment	1 week	Business Analyst
Literature Review & Industry Guidance	2 weeks	Research Analyst
Caltrans Procurement Analysis	1 week	Business Analyst
Performance-Based Framework Design	2 weeks	Project Manager & Transportation Consultant
Policy Recommendation Development	1 week	Entire Team
Documentation & Final Presentation	2 weeks	Entire Team

Total Estimated Project Duration: 10 weeks.

Table 2

Estimated Labor Hours

Activity	Hours
Planning	40

Current-State Assessment	40
Literature Review	80
Procurement Analysis	60
Framework Design	120
Policy Recommendation	60
Documentation	80
Presentation	40

Total Estimated Labor Effort: 520 Hours.

Table 3

Estimated Project Costs

Category	Estimated Cost
Labor	\$25,200
Software & Collaboration Tools	\$1,000
Office Supplies & Miscellaneous	\$800
Subtotal	\$27,000
10% Contingency Reserve	\$2,700
Total Estimated Budget	\$29,700

Note. These estimates are preliminary planning estimates and are expected to be refined as additional project information becomes available during project execution.

There is a contingency reserve of \$2,700 (10%) for unknown research needs, stakeholder-engagement activities or schedule changes that fall within the project scope. This money will not be used for scope increases outside of the approved project scope. The Project Lead can suggest contingency spending but won't be authorized until approved by sponsor. Approval will be noted

in project documentation. Any change proposed that impacts approved project scope, schedule, cost baseline, or approved deliverables will be subject to change control prior to the baseline changing.

Planning Assumptions

- A five-member consulting team will complete the project.
- Caltrans and government publications will be publicly available at no cost.
- No travel, field inspections, or contractor interviews are required.
- Existing Microsoft Office and collaboration software are available.
- The project is limited to research, framework design, and policy recommendations.
- Formal change control will be used if project scope changes.

All of these assumptions relate directly to the approved scope and research-based effort of the CFCEF. Because Caltrans will serve as the client, it is reasonable to assume that government information will be available for no direct cost to acquire because Caltrans has public resources to support procurement, strategic planning, and contracting (Caltrans, 2024c). Based on the project boundary defined in the scope statement, it is reasonable to assume there will be no travel or interviews. Therefore, the research will focus on publicly available government, industry, and scholarly resources. Given the scope and size of the team, it is reasonable to assume the research activities can occur concurrently with other phases of work, such as documentation, based on the schedule and associated WBS and labor-hour estimates.

Marchewka (2015) cautions that all estimates are based on stated planning assumptions and should be updated when better information is available. Therefore, the team will revisit these assumptions during weekly status meetings and at major milestones. If any of these assumptions are discovered to be invalid and will have material impacts on the approved baseline scope,

schedule, or cost, the team will utilize the project change- control process after recording the variance.

Overall, these estimates establish the schedule and cost baselines that will be used to monitor project performance during execution. Weekly status meetings will compare actual labor effort and work-package completion against the 520-hour labor baseline, 10-week schedule, and \$29,700 approved budget. Variances that threaten a milestone or require contingency funding will be documented and reviewed so corrective action can occur before later deliverables are affected. This creates a direct connection between the WBS, schedule, budget, and monitoring activities while preserving the approved project scope.

The planning estimates feed directly into the objective from Assignment 1 by allowing for the required resources to create a standardized contractor evaluation process for Caltrans. The new evaluation process will rank contractors based on performance, quality, safety, financial strength, schedule and project management instead of lowest cost. Creating realistic estimates will allow for a feasible baseline to finish creating the process while utilizing established project management techniques and procedures (California Department of Transportation, 2024; Marchewka, 2015). Proper planning also allows for one to be a good steward of their resources as described in Biblical scripture. Luke 14: 28 (New International Version) states that one should “look at the cost” before beginning a large project. By sitting down and estimating the cost to finish the work it will determine if one has the resources to complete the project.

2.4 High-Level Project Schedule (Jocelyne)

The project schedule establishes a 10-week timeline for completing the California Freeway Contractor Evaluation Framework (CFCEF) and aligns each activity with the work packages identified in the Work Breakdown Structure (WBS). The schedule also builds on the

time and cost estimates established in the previous section by identifying when project resources will be needed. Activities are sequenced based on their dependencies, while some research activities occur concurrently to make efficient use of the project timeline. Establishing this schedule baseline provides the team with a consistent reference for measuring progress throughout the Execution and Monitoring and Control phases.

Project Planning and Scope Definition begins in Week 1 and establishes the scope, responsibilities, and project management documentation needed for later activities. The Current-State Assessment begins during Week 2, while the Literature Review and Industry Guidance also begins in Week 2 and continues through Week 4. These activities can overlap because they use separate sources but contribute to the same research foundation. The Caltrans Procurement Analysis follows during Weeks 4 and 5 and depends on sufficient research being completed to identify the regulatory and procurement requirements that may affect the proposed framework.

Performance-Based Framework Design occurs during Weeks 5 through 7 and represents the project's most significant schedule dependency. Framework development relies on findings from the current-state assessment, literature review, and procurement analysis. Policy Recommendation Development follows during Week 8 because recommendations cannot be finalized until the framework's evaluation criteria and weighted scoring methodology have been developed. Documentation and Final Presentation activities conclude the project during Weeks 9 and 10 by integrating the completed project deliverables, conducting final reviews, and preparing the final presentation.

Table 4***Project Schedule***

Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10
Milestones		◆			◆		◆			◆
Planning & Scope										
Current-State Assessment										
Literature Review										
Procurement Analysis										
Framework Design										
Policy Recommendations										
Documentation & Final Proposal										

Note. The schedule illustrates the planned sequence and estimated duration of each work package across the 10-week project timeline. Milestones represent planned project phase completions and support schedule monitoring during weekly project status meetings.

The sequence of activities also identifies the project's primary dependency path.

Completion of the research and procurement analysis is necessary before the performance-based framework can be finalized, and the completed framework is required before the policy recommendation and final documentation can be completed. Delays affecting these dependent activities therefore present the greatest risk to the Week 10 completion date. Research activities with fewer dependencies may be adjusted or completed concurrently when necessary to protect the overall schedule.

The project schedule includes four major milestones: completion of planning and the current-state assessment by the end of Week 2, completion of research and procurement analysis by the end of Week 5, completion of the performance-based evaluation framework by the end of Week 7, and completion of final documentation and presentation by the end of Week 10. Progress will be measured against these milestones throughout the project. If an activity begins to fall behind, the team will first evaluate whether resources can be reassigned or concurrent work can be adjusted before considering changes to the approved baseline. Changes affecting project scope, the final deadline, or the approved cost baseline will require formal review rather than being absorbed informally.

2.5 Identify Risks and Mitigation Strategies (Jocelyne)

Identifying project risks during the planning phase allows the team to anticipate challenges that could affect the quality, feasibility, or timely completion of the contractor evaluation framework. Because this project focuses on research and policy development rather than implementation, the identified risks primarily relate to the quality of the framework and the team's ability to produce practical, evidence-based recommendations. Risk likelihood was evaluated using the project assumptions, available information, and professional judgment developed during the planning phase. Probability and impact are rated on a scale of 1 to 5 and multiplied to calculate a risk score, allowing higher-priority risks to receive greater monitoring and mitigation attention.

Table 5

Project Risk Assessment and Mitigation Strategies

Risk	Likelihood	Potential Impact	Mitigation Strategy	Probability	Impact	Risk Score
Limited publicly available contractor performance data	Moderate. The project relies primarily on publicly available information, which may not include every performance metric needed to support the proposed evaluation framework.	Some evaluation criteria may have limited supporting evidence, reducing the overall strength of the final recommendation.	Validate findings using multiple government publications, FHWA guidance, and peer-reviewed research while documenting any assumptions or data limitations.	3	4	12
Conflicting research findings	Moderate. Government publications and academic literature may recommend	Additional analysis may be required to justify the selected evaluation	Compare multiple credible sources, explain differences in the literature, and support	3	3	9

	different approaches to contractor evaluation and best-value procurement.	criteria and could increase project effort.	recommendations using the strongest available evidence.			
Existing procurement requirements may limit adoption of the proposed framework	Moderate. Some recommendations may not fully align with current Caltrans procurement policies or regulations.	Certain recommendations may require future policy or regulatory changes before they could be adopted.	Review existing procurement guidance throughout framework development and clearly distinguish between immediately feasible and long-term recommendations.	3	5	15
Evaluation framework becomes overly complex	Low. The framework will undergo multiple reviews during development, reducing the likelihood of unnecessary complexity.	An overly detailed framework could reduce usability and make consistent contractor evaluations more difficult.	Limit the framework to measurable performance criteria and use a clear, standardized scoring methodology.	2	4	8
Additional revisions identified during quality review	Moderate. Final reviews commonly identify improvements before project completion.	Additional revisions could affect the project schedule if significant changes are needed late in the project.	Conduct milestone reviews throughout the project and reserve limited schedule contingency for final revisions.	3	3	9

Note. Risk assessments represent the Planning baseline. Probability and impact are rated from 1 (low) to 5 (high), and the risk score is calculated by multiplying probability by impact.

Project risks will be reviewed at each planned project milestone to determine whether likelihood or potential impacts have changed as additional information becomes available.

Ongoing risk reviews also help identify emerging uncertainties that may affect the project schedule, allowing the team to evaluate potential impacts and implement appropriate response

strategies before significant schedule deviations occur (Acebes et al., 2021). Updating the risk register throughout the project helps the team respond proactively while maintaining alignment with the approved scope, schedule, and overall project objectives. Risks that materialize will be transferred to the issue log for active resolution during Monitoring and Control.

2.6 Create a Resource Plan (Herman)

Effective resource planning ensures that appropriate personnel, financial resources, equipment, software, and project supplies are available throughout the project's life cycle. Resource planning supports efficient allocation of responsibilities, minimizes schedule delays, and helps ensure project objectives are achieved within the approved scope, schedule, and budget (Project Management Institute [PMI], 2025). Because the California Freeway Contractor Evaluation Framework (CFCEF) project is a research and policy development initiative, the primary resources consist of skilled personnel supported by commonly available technology and office resources.

Human Resources

The project will be completed by a five-member consulting team with clearly defined roles and responsibilities. The Project Manager will oversee project planning, scheduling, stakeholder communication, and overall project coordination while monitoring project performance. Two Business Analysts will evaluate Caltrans' existing contractor selection process, analyze procurement regulations, and assist in developing the contractor evaluation criteria. A Research Analyst will conduct the literature review, analyze scholarly and government publications, and ensure recommendations are supported by current evidence. A Transportation Consultant will provide subject-matter expertise in transportation procurement,

construction contracting, and contractor performance evaluation. During the final project phase, all team members will participate in developing policy recommendations, quality assurance reviews, and preparation of the final report and presentation.

Financial Resources

The resource plan aligns with the project's estimated budget developed during the planning phase. Labor remains the largest project expense because the majority of work consists of research, analysis, framework development, documentation, and stakeholder review. The estimated labor cost is \$25,200, supported by \$1,000 allocated for software licensing and collaboration tools and \$800 designated for office supplies and miscellaneous project expenses. These costs produce a subtotal of \$27,000. A 10% contingency reserve (\$2,700) increases the total project budget to \$29,700, allowing the project team to address unforeseen research requirements, additional stakeholder requests, schedule adjustments, or minor scope changes without significantly affecting project performance.

Technical Resources

The project requires standard business technology rather than specialized engineering equipment. Each team member will use a laptop or desktop computer with reliable high-speed internet access to conduct research and collaborate remotely. Microsoft Word will be used to prepare project documentation; Microsoft Excel will support cost estimation, scheduling, and data analysis, and Microsoft PowerPoint will be used to develop the final presentation. Microsoft Teams and Outlook will facilitate virtual meetings, project communication, document sharing, and task coordination. Scholarly databases, including Google Scholar and the California Baptist

University Library, along with publicly available resources from the California Department of Transportation (Caltrans), the Federal Highway Administration (FHWA), and the U.S. Department of Transportation, will provide the research needed to develop the proposed contractor evaluation framework.

Project Supplies and Equipment

Although this project does not require construction materials or field equipment, several office and administrative supplies are necessary to support project execution. Required supplies include:

- Laptop or desktop computer for each team member
- High-speed internet connection
- Microsoft 365 (Word, Excel, PowerPoint, Outlook, and Teams)
- Project scheduling software (Microsoft Project or equivalent)
- Shared cloud storage using Microsoft OneDrive or SharePoint
- Access to Google Scholar, the California Baptist University Library, and government databases
- Printer and scanner for document review when needed
- Printer paper, notebooks, binders, folders, sticky notes, and writing materials for project documentation and meeting notes
- Video conferencing equipment (webcam, headset, and microphone) for virtual collaboration and presentations

These resources support research, communication, document management, scheduling, and collaboration while keeping project costs within the approved budget.

Resource Allocation Strategy

Resources will be assigned according to the Work Breakdown Structure (WBS) and the ten-week project schedule. During Weeks 1 through 5, the majority of effort will focus on planning activities, literature review, procurement analysis, and current-state assessment, requiring greater involvement from the Research Analyst and Business Analysts. During Weeks 6 through 8, emphasis shifts toward development of the California Freeway Contractor Evaluation Framework and preparation of policy recommendations, with increased collaboration among the Project Manager, Transportation Consultant, and Business Analysts. During the final two weeks, all project team members will participate in documentation, quality assurance, editing, and preparation of the final presentation. Weekly project status meetings will allow the Project Manager to monitor workload, resolve resource conflicts, and redistribute assignments when necessary to maintain the approved project schedule and budget (Kerzner, 2025).

The proposed resource plan aligns directly with the project scope, schedule, and estimated budget while ensuring that the personnel, technology, supplies, and financial resources necessary to complete the California Freeway Contractor Evaluation Framework are available throughout the project. Effective resource planning provides a foundation for successful project execution by supporting collaboration, efficient resource utilization, and timely delivery of all project deliverables

Execution Phase

3.1 Team Roles & Responsibilities (Ezeoma Igwe)

During the Execution phase, responsibilities are assigned according to the RACI matrix established during the Planning phase. Assigning a lead for each major work area establishes

clear accountability, minimizes duplicated effort, and ensures execution activities remain aligned with the project's communication, documentation, and delivery requirements.

- **Ezeoma Igwe – Project Lead:** Coordinates Weekly Project Leadership Meetings (WPLMs), consolidates team contributions, assigns and monitors tasks, and ensures execution remains aligned with the approved project scope, schedule, and objectives.
- **Pujan Thanki – Communication Lead:** Oversees the Communication Plan, including communication tools, meeting protocols, documentation standards, and escalation procedures.
- **Donavin Tuiteleapaga – Stakeholder Engagement Lead:** Manages the Stakeholder Engagement Strategy, including the California Freeway Contractor Evaluation Framework (CFCEF) Project Page, stakeholder feedback survey, and stakeholder engagement log.
- **Jocelyne Olivares – Implementation Milestones Lead:** Defines and monitors the Implementation Milestones, ensuring deliverables are completed in sequence and within the project's 10-week schedule.
- **Herman Steele – Ethics & Compliance Lead:** Oversees Ethical and Cross-Cultural Considerations, including fairness in evaluation criteria, inclusive stakeholder engagement, and compliance with data privacy and conflict-of-interest requirements.

Table 6

Team Roles and RACI Summary

Team Member	Role	Primary Deliverables
Ezeoma Igwe	Project Lead	WPLM coordination. Task

		assignment, scope alignment
Pujan Thanki	Communication Lead	Communication Plan
Donavin Tuiteleapaga	Stakeholder Engagement Lead	Stakeholder Engagement Strategy
Jocelyne Olivares	Implementation Milestones Lead	Implementation Milestones
Herman Steele	Ethics & Compliance Lead	Ethical/Cross-Cultural Considerations

Finally, this is consistent with the biblical principle of delegated leadership described in Exodus 18:17–23, where Jethro counsels Moses to appoint capable leaders over specific matters so that only the most significant issues require his direct involvement, the team has structured its Execution-phase rolls around distinct areas of ownership. This approach allows the Project Led to focus on overall coordination and escalation, while each team member exercises full ownership and accountability over their assigned deliverable.

3.2 Communication Plan (Pujan)

Effective communication integrates every area of project management, and its absence is one of the most common causes of project failure. Zulch (2014) found that communication is not merely one knowledge area among many, but the main factor that brings cost, scope, and time together, and that poor communication during a project directly damages schedule, cost, worker safety, and quality.

For the California Freeway Contractor Evaluation Framework (CFCEF) project, this risk is elevated for two reasons. First, the project team operates through virtual collaboration tools rather than a shared physical office. Second, the deliverable is a policy framework rather than a physical asset, meaning that nearly every project output is a document that must be drafted, reviewed, revised, and approved by multiple parties. In this environment, communication failures do not appear as visible construction defects, but instead they appear as conflicting draft versions, duplicated effort, and decisions made without the team's awareness.

This communication plan is a component of the overall project management plan and is intended to be a living document that is revised as the project progresses (California Department of Transportation, 2007). Its purpose is to answer four questions for every project deliverable: who needs the information, when they need it, who delivers it, and how it is delivered (Caltrans, 2007). The plan below addresses internal communication among the project team and the routine reporting with the project sponsor.

Communication Tools and Platforms

The project team will standardize a limited set of tools so that information has one predictable location rather than being scattered across multiple channels. Microsoft teams will serve as the primary platform for both scheduled meetings and day-to-day coordination, with dedicated channels created for each major workstream of the framework, such as evaluation criteria development, research and benchmarking, stakeholder feedback, and document review. Email will be reserved for formal response, including deliverable submissions, approval requests, sponsor reporting, and any communication that must be preserved as a formal project record.

A SharePoint document library will accommodate all working drafts of the framework, meetings, the action item log, and the stakeholder engagement log, ensuring the team works from a single authoritative version of each document rather than emailing attachments back and forth.

Meeting Protocols and Documentation Standards

Meetings will follow a consistent protocol so that they produce decisions rather than consuming time. An agenda will be distributed at least twenty-four hours before each meeting, and minutes will be posted to SharePoint within twenty-four hours. Every minute set will record attendance, decisions reached, and action items with a named owner and a due date, since an action item without an owner is functionally unassigned.

Project records will be retained and archived consistently with Caltrans practice, which treats standardized filing as a safeguard against the most error-prone form of project communication: communication across long spans of time, where a future recipient must understand a decision after the original author is no longer available (Caltrans, 2007).

Escalation and Conflict Resolution

Disagreements over evaluation criteria, scoring weights, or scope are expected on a project of this nature, and the plan assumes they will occur rather than treating them as expectations. Consistent with Caltrans conflict management practice, issues will be resolved at the lowest possible level and escalated on a defined timetable (Caltrans, 2007). At the first level, the affected task leads, and the project manager attempts resolution within one to fifteen days. If any resistance remains, the issue is escalated to the project sponsor for a decision within sixteen to thirty days.

Issues that cannot be resolved at that level, such as those requiring a change to approved scope or budget, are escalated through the project's change control process. Each escalated issue will be logged with the date, the parties involved, the positions taken, and the resulting decision, so the rationale behind contested framework provisions remains traceable.

Monitoring Communication Effectiveness

The communication plan will be evaluated during execution rather than assumed to be working. Caltrans recommends periodically soliciting feedback from project participants on five specific dimensions: the type of information received, its frequency, its depth and level of detail, its format, and the method of transmittal (Caltrans, 2007). The project manager will conduct this review at each milestone gate and adjust the plan where a participant reports receiving too much detail, too little, or information arriving too late to act on.

This review matters because communication problems in virtual teams are not solved simply by communicating more. Marlow et al. (2017) concluded that communication quality exerts a stronger influence on virtual team performance than communication frequency, meaning that a team can hold frequent meetings and still fail if the information exchanged is unclear, incomplete, or badly timed. The CFCEF team will therefore treat clarity and timeliness as the measures of success for this plan, not the number of meetings held.

3.3 Stakeholder Engagement Strategies (Donavin)

Extensive stakeholder management will remain significant during the project execution phase of the California Freeway Contractor Evaluation Framework project. Because the purpose of this project is to develop a standardized contractor evaluation framework to improve Caltrans' current process of contractor selection, it will be important for members of the project team to

continue to meet with stakeholders. Meeting with stakeholders will allow the project team to design a framework that meets the needs of individuals who will administer the project, individuals involved in the process, and those who benefit from California’s transportation procurement process. Maintaining open communication with stakeholders will also give the project team an opportunity to fact-check research, gather recommendations, and adjust initiatives to ensure that constructive feedback is implemented before project completion (Project Management Institute [PMI], 2021).

To improve transparency and stakeholder communication during the execution phase of the project, it is recommended that the project team create a California Freeway Contractor Evaluation Framework Project Page located on Caltrans’ current website. The project page would serve as a communication hub during development and would provide stakeholders with current information about the project’s progression, upcoming milestones, meeting agendas, copies of draft frameworks, and opportunities for feedback. By utilizing Caltrans’ current website to display the project page, stakeholders will be able to hear from the project team through a platform they are familiar with and increase transparency with contractors, government officials, and California citizens (California Department of Transportation [Caltrans], 2024).

Table 7
Stakeholder Engagement and Communication Plan

Stakeholder	Communication Method	Frequency	Purpose
Caltrans (Project Sponsor)	Weekly meetings, Microsoft Teams, email, and project	Weekly	Review project progress, validate framework components, and

	webpage		approve major deliverables.
Contractors and Construction Firms	Surveys, webinars, project webpage, and virtual workshops	At major milestones	Gather recommendations regarding evaluation criteria, procurement transparency, and proposed framework practicality.
California Residents, Commuters, and Taxpayers	Public project webpage, online surveys, and public comment forms	Midpoint and final review	Collect feedback regarding previous Caltrans construction projects and identify contractor performance characteristics most important to the public.
Federal Highway Administration (FHWA)	Email correspondence, virtual meetings, and document reviews	Monthly	Verify that the proposed framework remains consistent with federal procurement guidance and best-value contracting practices.
Project Team	Weekly Project Leadership Meetings (WPLMs), Microsoft Teams, shared documents	Weekly	Coordinate project activities, monitor risks, assign responsibilities, and review stakeholder feedback.

Project Communication Strategy

During development, stakeholders will be able to track progress of the project through the CFCEF Project Page. Project goals, project development timeline, meeting agendas, framework drafts, FAQs, and progress updates will be available on the webpage for reference. Additionally, stakeholders can pose questions and make suggestions via the webpage for the project team to evaluate. This will allow open communication to be sustained during the framework development process.

Stakeholder Feedback Strategy

One major goal of stakeholder engagement will be to gather information that will validate the proposed contractor evaluation framework. A Stakeholder Feedback Survey will be posted on the project website to solicit feedback from Caltrans employees, contractors, and the public on contractor selection and performance. Members of the public will be encouraged to provide their thoughts on past Caltrans projects they have experienced. Feedback will include rating of contractor performance on construction quality, traffic management, work-zone safety, meeting of schedule, communication with the public, and overall satisfaction with the project. Open-ended comments will also be available for stakeholders to list frequent concerns or suggestions on things to look for when evaluating future contractors.

Contractors and construction companies will be able to provide feedback on the procurement process including transparency, request for bid (RFB) requirements, evaluation factors, and implementation difficulties. Caltrans employees will review sections of the draft framework and provide feedback based on their knowledge of administering procurement processes and managing transportation projects. This feedback will allow the project team to collect qualitative data that supports the research completed during the Planning Phase and confirm the framework is meeting the needs of key stakeholders.

Stakeholder Feedback Review

During execution, comments from stakeholders will be tracked and assessed to identify if changes to the CFCEF are warranted. Comments received from stakeholder meetings, surveys, workshops, email, or the project website will be logged into a stakeholder engagement log. The log will identify who the stakeholder group is, when it was received, what the recommendation is, which area of the project it relates to, who in the team is responsible for reviewing it, and the disposition of the recommendation. Tracking stakeholder engagement in a central log allows for accountability while ensuring all feedback is consistently addressed during project execution (PMI, 2021).

During weekly project team meetings (WPLMs), the team will review new stakeholder feedback and look for trends across common concerns, themes, and suggestions to improve the proposed evaluation framework. Feedback from Caltrans, contractors, and California residents will be given extra consideration since stakeholders from these groups will be responsible for administering, working under, and benefiting from the finalized procurement process. Looking for trends versus individual opinions will allow the team to make decisions that are objective and supported by more than one stakeholder. Feedback that will improve the evaluation framework and does not deviate from the project's approved scope and objectives will be considered for Implementation in future framework drafts. Recommendations that change the project scope, schedule, cost, or deliverables will be processed through the project's change control process before being considered for implementation. This allows stakeholder feedback to be heard and

considered while preventing scope creep and ensuring the project still aligns with the project management plan approved by the customer (PMI, 2021).

At the end of each review cycle, stakeholders will be updated on major decisions made by the project team through the project website and scheduled progress reports. When feasible, the project team will publish summaries of stakeholder feedback received and resulting changes made to the project so stakeholders can see how their input was considered and if it had any impact on framework development. This allows stakeholders to see they played an important role in project development, which can encourage future participation and foster trust between stakeholders and the project team. By including stakeholders in the project development process, the project team is applying Proverbs 15: 22 from the Bible (New International Version): “Plans fail for lack of counsel, but with many advisers they succeed.”

Stakeholder engagement will also be evaluated with measurable indicators so the project team can determine whether the engagement strategy is working rather than relying only on the number of communication activities completed. The project will target participation from at least 90% of the stakeholder groups and planned engagement opportunities identified in the engagement plan, document a formal disposition for 100% of stakeholder recommendations received, and acknowledge submitted questions or comments within five business days when contact information is available. At milestone reviews, the Stakeholder Engagement Lead will report these results to the project team and identify any stakeholder group with limited participation. If participation falls below the target, the team may increase webpage notices, survey outreach, or virtual engagement opportunities without changing the approved project scope. These measures connect the Execution Phase engagement strategy directly to the

Monitoring and Control Phase KPIs and provide objective evidence of whether stakeholder involvement is supporting framework development.

Monitoring Stakeholder Communication Effectiveness

Project team members will measure if stakeholder communication channels are reaching targeted audiences and generating actionable feedback instead of assuming communication is successful because information was sent out. The Stakeholder Engagement Lead will analyze project-page views, survey participation rates, attendance at stakeholder meetings, number of meaningful comments and percent of stakeholder recommendations with a recorded disposition at each WPLM. The team will also assess if stakeholders indicate information is being shared with adequate clarity, detail, frequency, and timeliness (California Department of Transportation, 2007). Changes will be made to the communication plan if participation does not meet project benchmarks or repetitive questions suggest project information is not clearly presented. Adjustments may include revising webpage content, rephrasing survey questions, increasing outreach to specific stakeholders, or adjusting when project updates are distributed. Findings will be presented at WPLMs and tracked in the stakeholder engagement log to ensure a documented history of communication improvements. This process will link stakeholder engagement efforts to the project's defined KPIs and allow for continuous improvement during project execution

3.4 Implementation Milestones (Jocelyne)

Implementation milestones provide structured checkpoints for evaluating whether the CFCEF project is ready to advance to the next stage of work. The four milestones established during Planning occur at Weeks 2, 5, 7, and 10 and correspond with completion of major project deliverables. During Execution, these milestones serve as formal review points where the team

evaluates completed work, resolves outstanding concerns, and determines whether project activities can proceed as scheduled.

The Week 2 milestone confirms completion of project planning and the current-state assessment. The team reviews the project scope, schedule, responsibilities, and assessment findings before continuing with the remaining research activities. A Go decision requires the planning baseline and current-state assessment to be complete and accepted. A No-Go decision requires unresolved planning or assessment issues to be corrected before dependent work proceeds.

The Week 5 milestone confirms completion of the literature review and Caltrans procurement analysis. These findings provide the evidence and regulatory foundation required for development of the performance-based evaluation framework. A Go decision requires sufficient research support and identification of applicable procurement constraints. Significant evidence gaps or unresolved regulatory questions result in a No-Go decision until they are addressed.

The Week 7 milestone evaluates completion of the CFCEF performance-based framework, including its evaluation criteria and weighted scoring methodology. Because the policy recommendation depends directly on the completed framework, the team reviews the framework for clarity, evidence support, usability, and alignment with project scope before proceeding. A Go decision authorizes development of the final policy recommendation, while major unresolved framework issues require revision before advancement.

The Week 10 milestone confirms completion of the final documentation and presentation. The team verifies that required deliverables are complete, quality reviews have been addressed,

and the final CFCEF and policy recommendation remain consistent with the approved project objectives and scope. A Go decision represents final project acceptance, while unresolved major findings require correction before project closeout.

3.5 Ethical or Cross-Cultural Situations (Herman)

Because the California Freeway Contractor Evaluation Framework (CFCEF) will shape how public construction dollars are awarded across the state, the project team must execute the project in a way that is fair, inclusive, and transparent to every group affected by Caltrans' procurement process. Ethical and cross-cultural considerations are addressed throughout execution rather than treated as a single checkpoint, since fairness concerns can surface at any point during framework development, stakeholder engagement, or pilot testing (PMI, 2021).

Fairness and Inclusivity in Evaluation Criteria

A central ethical risk of the CFCEF is that its evaluation criteria could unintentionally favor large, established contractors and disadvantage small, minority-owned, women-owned, or disabled veteran business enterprises (DBEs) that have historically competed for Caltrans contracts. To guard against this, the project team will review each proposed scoring criterion for disparate impact before it is finalized and will consult Caltrans' existing DBE and civil rights program staff during the review process. Where a criterion could disadvantage smaller or historically underrepresented firms without a legitimate performance justification, the team will revise or reweight it. The team will also disclose the full scoring methodology on the CFCEF Project Page so that all contractors, regardless of size or resources, compete under the same known standard (Caltrans, 2024).

Cross-Cultural Accessibility of Stakeholder Engagement

Because California residents, commuters, and taxpayers are among the CFCEF's key stakeholders, the project team recognizes that this population spans a wide range of languages, cultural backgrounds, and levels of comfort with government processes. If public engagement is limited to English-language surveys and webpage content, the team risks excluding non-English-speaking communities from providing input on contractor performance and procurement transparency, even though these communities are directly affected by freeway construction projects in their neighborhoods. To address this, the CFCEF Project Page and public comment forms will be made available in the languages most commonly spoken by Caltrans' constituent communities, and public workshops will offer interpretation upon request. The team will also ensure that the webpage and survey tools meet Americans with Disabilities Act (ADA) accessibility standards so that residents with visual, auditory, or motor impairments can participate on equal footing with other stakeholders (Caltrans, 2024).

Data Privacy and Conflict of Interest

The stakeholder feedback surveys and public comment forms described in Section 3.3 will collect opinions and, in some cases, personal information from Caltrans employees, contractors, and members of the public. The project team will limit data collection to what is necessary for framework validation, store responses securely, and anonymize public survey results before they are published or summarized on the project website. Contractor-submitted feedback regarding the evaluation criteria will similarly be treated as confidential until aggregated, so that no single firm's comments can be tied back to that firm in a way that could affect its standing under the new framework. In addition, any project team member or Caltrans reviewer with a financial or personal relationship to a contracting firm involved in the pilot testing phase will be required to disclose that relationship and recuse themselves from scoring or

feedback decisions related to that firm. These safeguards are intended to protect stakeholder privacy, preserve the integrity of the evaluation process, and maintain public trust in a framework that is ultimately meant to make Caltrans' contractor selection process more transparent and equitable (PMI, 2021).

Monitoring and Control Phase

4.1. Establish Performance Tracking Tools (Herman)

For the California Freeway Contractor Evaluation Framework (CFCEF) project, performance tracking tools are selected to monitor progress against the schedule, budget, and scope of baselines established during the Planning phase. Because this project produces research and policy deliverables rather than physical construction work, tracking tools are adapted to measure completion of analytical work packages and documentation milestones rather than physical units of work in place.

Earned Value Management (EVM)

Earned Value Management will be used to objectively measure schedule and cost performance against the baselines established during planning (520 labor hours; \$27,000 subtotal plus a 10% contingency reserve, for a total approved budget of \$29,700). Because the project's "cost" is primarily labor hours rather than purchased materials, EVM will track percent-complete of each WBS work package (e.g., Literature Review, Procurement Analysis, Framework Design) against its planned effort allocation:

- **Planned Value (PV):** Budgeted labor hours scheduled to be expended by a given week, based on the 10-week schedule baseline

- Earned Value (EV): Budgeted hours for work packages completed (e.g., if Current-State Assessment, budgeted at 40 hours, is finished on schedule, its full 40 hours are "earned")
- Actual Cost (AC): Labor hours logged by the team for that work
- Schedule Performance Index (SPI) = EV / PV and Cost Performance Index (CPI) = EV / AC — used to flag whether a work package is falling behind schedule or consuming more effort than budgeted, in time to correct course before it affects a downstream milestone (Mayo-Alvarez et al., 2022)
- EVM metrics will be calculated at each of the four schedule milestones (end of Weeks 2, 5, 7, and 10) rather than continuously, matching the project's existing milestone structure.

Budget Tracking

A budget-vs-actual tracker will be maintained in Excel, broken down by the three cost categories established in the Planning phase: labor (\$25,200), software/collaboration tools (\$1,000), and office supplies/miscellaneous (\$800). The Project Lead will monitor how much of the 10% contingency reserve (\$2,700) has been drawn down, since the reserve is the mechanism, the team already designated for absorbing unforeseen research needs or minor scope adjustments without a formal change request.

Progress Reports

We will conduct weekly meetings to check progress and make changes as needed. Communication will be done by email and teams' meetings as needed as well as progress meeting with shareholders for updates as needed

- Work packages completed vs. scheduled for that week
- Labor hours logged vs. budgeted (feeding the EVM calculation)
- Progress toward the next milestone (Weeks 2, 5, 7, or 10)
- Any risks or issues newly identified or requiring escalation

Milestone Tracking

The four schedule milestones already defined in the Planning phase (planning/current-state assessment by Week 2, research/procurement analysis by Week 5, evaluation framework by Week 7, final documentation/presentation by Week 10) will serve as the primary checkpoints for go/no-go progress review, consistent with Table 4's schedule baseline.

Table 8

Summary

Tool	What It Measures	Frequency	Owner
EVM (SPI/CPI)	Schedule & effort performance vs. baseline	At each milestone (Wk 2, 5, 7, 10)	Project Lead
Budget tracker	Actual spend/hours vs. \$29,700 budget	Weekly	Project Lead
Progression reports	Task completion, blockers, risks	Weekly	All team members
Milestone checkpoint	Deliverable completion vs. schedule	Wk. 2, 5, 7, 10	All team members

4.2 Update Risk and Issue Logs (Jocelyne)

The risk register established during Planning will be updated throughout Monitoring and Control to reflect changes in probability, impact, risk score, response actions, and status. Each risk will also be assigned an owner responsible for monitoring the risk and documenting changes. Risks with higher scores will receive greater attention, particularly when they could affect the project's scope, schedule, quality, or ability to produce a feasible recommendation. Risk reviews will occur during weekly project status meetings and at the four project milestones in Weeks 2, 5, 7, and 10.

The Planning risk register identified five primary risks: limited publicly available contractor performance data, conflicting research findings, procurement requirements that may limit adoption, excessive framework complexity, and additional revisions identified during quality review. As the project progresses, the team will reassess the probability and impact of each risk using the same 1-to-5 scale established during Planning. The resulting risk score will be compared with the original Planning baseline to determine whether mitigation actions should be continued, modified, or escalated.

The Issue Log will be used only when a monitored risk materializes or another active project problem is identified. Because no specific project issues have been documented at this stage, the log will remain available for future use rather than including assumed entries. Once an issue is identified, the project team will document its impact, assign responsibility for resolution, establish a target completion date, and monitor corrective actions until the issue has been resolved (Project Management Institute [PMI], 2021). Issues affecting approved scope, schedule, cost, or major deliverables will be escalated to the Project Lead and addressed through the established change-control process when necessary.

To ensure that risks and issues remain actively managed, open entries will be reviewed at least every 14 days. High-severity risks without a status update for 14 days will require follow-up from the assigned owner. This review cycle supports the project's open risk aging metric and provides a measurable method for determining whether identified risks are being actively monitored rather than simply recorded. Maintaining current risk and issue logs also provides the project team with documentation of mitigation decisions, issue resolution, and any resulting changes throughout the remainder of the project.

Rather than creating a separate risk register, the Project Risk Assessment and Mitigation Strategies established during Planning will serve as the project's baseline risk log. During Monitoring and Control, the existing register will be updated by assigning a responsible owner to each risk and adding current status, last review date, and target review date fields. Probability, impact, and risk scores will also be reassessed when project conditions change.

4.3 Define Quality Control Mechanisms (Ezeoma)

Because the CFCEF project produces analytical models, research syntheses, and policy recommendations rather than physical construction deliverables, quality control mechanisms focus on analytical rigor, stakeholder credibility, regulatory alignment, and documentation standards (Marchewka, 2015; PMI, 2021). Quality control is not treated as a single end-of-project inspection; rather, it operates continuously throughout project execution to prevent minor errors from compounding into major project failures (Busby, 2001; PMI, 2021). The project team will apply the following process to each of the project deliverables before it is completed.

1. **Peer Review and Analytical Audits:** Before each milestone review, team members conduct cross-peer inspections of one another's work to identify errors, logic gaps, and

unsupported assumptions. In addition, traceability checks ensure that 100% of the proposed CFCEF evaluation criteria and scoring weights are supported by primary research or established procurement guidance, providing a clear audit trail for every recommendation.

2. **Quality Gateways and Action Thresholds:** Quality control metrics track the number and severity of substantive defects or revision requests raised during review cycles (Cheung et al., 2004), with the expectation that defect density decreases over successive revisions. To maintain quality, no more than 20% of deliverables should require major rework after review. If the rework rate exceeds 35%, the Project Lead initiates a formal review to reassess project requirements, resource allocation, and work assignments.
3. **Format, Standards, and Accessibility Verification:** All deliverables are reviewed for compliance with academic and professional standards, including APA formatting, consistent terminology, and clear document structure. Public-facing materials are also verified for ADA accessibility and multilingual availability to promote equitable access for all California residents (Caltrans, 2024), and inclusive stakeholder participation.

Deliverable Review Process

Deliverables undergo a structured, multi-stage review at each of the four project milestones (Weeks 2, 5, 7, and 10) to verify quality, completeness, and alignment with project objectives before progressing to the next phase.

Figure 1

CFCEF Deliverable Quality Review Process



Note: The figure illustrates the four-stage review sequence applied to each project deliverable, from internal team review through final disposition and sign-off.

Stage 1: Internal Team Review: The team member who did not author the deliverable reviews for source traceability, internal consistency with prior deliverables, and adherence to documentation standards. Comments are logged and must be resolved before the draft advances. To maintain strong governance during peer reviews, technical or editorial disagreements regarding evaluation criteria, scoring weights, or analytical interpretations will follow a defined internal resolution protocol:

- **Step 1 (Peer Consensus & Majority Vote):** Reviewers and task leads will first attempt to resolve differences through direct discussion during weekly team meetings. If consensus cannot be reached, the issue will be put to a majority vote among the five consulting team members.
- **Step 2 (Project Lead Decision):** If a vote results in a tie or if the issue directly impacts overall project scope, schedule, or budget baselines, the Project Lead holds final decision-making authority to determine the path forward.

Stage 2: Milestone Gate Review: At each of the four project milestones (Weeks 2, 5, 7, 10), the full team reviews the deliverable together to confirm it's sufficient to support the next phase of work, not just complete in isolation.

Stage 3: Stakeholder & Sponsor Review: The draft is shared with Caltrans and relevant stakeholders via the CFCEF Project Page and email. Comments are logged in the stakeholder engagement log, with sponsor comments flagged as requiring resolution before proceeding.

Stage 4: Formal Disposition & Sign-off: The Project Lead records the final disposition of all review comments (accepted, deferred, or rejected with rationale) and formally closes out the deliverable as complete, or returns it for revision if it failed review.

Finally, the quality mechanism control for this project is supported by Biblical Principles. The project's quality control approach reflects biblical principles of verification, integrity, and excellence. *1 Thessalonians 5:21* states "test everything; hold fast what is good" emphasizes testing and verifying findings before acceptance, supporting evidence-based recommendations. *Proverbs 16:11* states, "Honest scales and balances belong to the Lord; all the weights in the bag are of his making" which underscores honesty and fairness by ensuring objective and transparent evaluation criteria. *Colossians 3:23* encourages excellence and faithful stewardship, reinforcing the team's commitment to delivering high-quality work for Caltrans and California taxpayers.

4.4 Evaluation Metrics (Pujan)

This section defines the specific measures those instruments will produce. A distinction is made here between evaluation metrics and key performance indicators. Kerzner (2017) explains that a metric is a measurement of a specific attribute of project work, while a key performance indicator is a metric elevated to the status of a success criterion because it directly reflects

whether the project is achieving its intended outcome. Every KPI is a metric, but most metrics are not KPIs. The evaluation metrics defined below are operational and leading in nature.

Criteria for Metric Selection

The project team applied four criteria in selecting the metrics below. First, each metric must be measurable from data the team is already generating through the tracking tools, so that measurement does not become an additional workload on a project. Second, each metric must have a defined target and a defined threshold at which corrective action is engaged, since a measurement with no threshold produces observation rather than control. Third, each metric must be actionable, meaning the team can identify a specific response when the metric moves out of tolerance. Fourth, the metric set must cover more than cost and schedule.

Cheung et al. (2004) found that project performance systems relying primarily on financial parameters suffer from clear inadequacies, including a reliance on lagging measures and a failure to capture quality, stakeholder relationships, and communication effectiveness. The CFCEF deliverable is a policy framework whose value depends on stakeholder credibility and analytical rigor.

Evaluation Metrics for the CFCEF Project

The metrics are organized into five categories: schedule, effort and cost, scope and deliverable progress, quality, and stakeholder engagement.

Table 9

CFCEF Evaluation Metrics

Metric	Measurement Method	Target	Action Threshold	Frequency / Owner
<i>Schedule Performance</i>				
Schedule Performance Index (SPI)	$EV \div PV$, from the EVM calculation in Section 4.1	$SPI \geq 0.95$	$SPI < 0.90$ triggers a schedule recovery discussion	Milestones (Week 2, 5, 7, 10) / Project Lead
Milestone adherence	Milestones met on or before baseline date $\div$ milestones scheduled to date	100%	Any milestone missed by more than 3 days	Milestones / Project Lead
Weekly task completion rate	Work packages completed $\div$ work packages scheduled that week	$\geq 90\%$	Two consecutive weeks below 80%	Weekly / All members
<i>Effort and Cost</i>				
Cost Performance Index (CPI)	$EV \div AC$, from the EVM calculation in Section 4.1	$CPI \geq 0.95$	$CPI < 0.90$	Milestones / Project Lead
Labor hour variance	Hours logged $\div$ hours budgeted, by work package	Within $\pm 10\%$ of the 520-hour baseline	Any work package exceeding budgeted hours by 20%	Weekly / Project Lead
Contingency reserve consumption	Reserve drawn $\div$ \$2,700 approved reserve	$\leq 50\%$ consumed by Week 7	More than 75% consumed before Week 8	Weekly / Project Lead
<i>Scope and Deliverables</i>				
Deliverable completion rate	Deliverables completed $\div$ deliverables scheduled to date per the WBS	100% by Week 10	Any deliverable not started by its scheduled start date	Weekly / All members
Change request volume	Count of formal change requests raised and their disposition status	All requests dispositioned within 5 business days	Any request open more than 10 business days	Weekly / Project Lead
<i>Quality</i>				

Metric	Measurement Method	Target	Action Threshold	Frequency / Owner
Review defect density	Substantive comments raised per framework section during peer and sponsor review	Declining trend across successive drafts	An increase in defects between two consecutive drafts of the same section	Per review cycle / Reviewers
Rework rate	Deliverables requiring major revision after review ÷ deliverables reviewed	≤ 20%	Above 35%	Per review cycle / Project Lead
Evaluation criteria traceability	Proposed scoring criteria supported by a documented research or regulatory source	100%	Any criterion advancing to a draft without a documented source	Milestones / All members
<i>Stakeholder Engagement</i>				
Stakeholder response rate	Responses received ÷ stakeholder groups solicited, per engagement cycle	≥ 75% of identified groups	Any group unresponsive across two consecutive cycles	Per cycle / All members
Feedback disposition turnaround	Median days from feedback receipt to logged disposition in the stakeholder engagement log	≤ 7 days	Median exceeding 14 days	Weekly / Project Lead
<i>Risk</i>				
Open risk aging	Median days a risk remains open in the risk log without a status update	≤ 14 days	Any high-severity risk without an update for 14 days	Weekly / Risk owner

Note. EV = earned value; PV = planned value; AC = actual cost; WBS = work breakdown structure. Metrics reported as at risk fall between the target and the action threshold and are addressed at the weekly progress meeting. Metrics breaching the action threshold are escalated to Project Lead.

Variance Threshold and Corrective Action

Metrics are only useful if the team has agreed in advance what constitutes an acceptable deviation. The project team will apply a three-level tolerance to each metric. Values within target are reported as on track and require no action. Values falling between the target and the corrective action threshold are reported as at risk and are discussed at the weekly progress meeting, where the team member responsible proposes a recovery plan. Values breaching the corrective action threshold are escalated to Project Lead, who determines whether the situation can be absorbed within the contingency reserve or whether it requires a formal change request. Mayo-Alvarez et al. (2022) note that baseline performance management functions by comparing progress against the scope, schedule, and cost baseline established during planning, which means the value of these metrics depends entirely on the baseline remaining a stable reference point.

Relationship with Project Reporting

These metrics will be compiled into a single one-page performance summary distributed with the weekly progress report, showing each metric's current value, its target, its tolerance status, and the direction of movement since the prior week. Kerzner (2017) emphasizes that the value of a metrics program lies in the quality of the reporting rather than the volume of data collected, and that presenting too many measures eliminates the few that require action. The summary will lead with any metric currently outside tolerance rather than presenting all fourteen measures at equal weight. At each milestone, the accumulated metric feeds the KPI assessment which evaluates whether the project is achieving its intended outcomes rather than simply executing on schedule. This approach reflects the guidance in the PMBOK guide that monitoring and controlling is a continuous process performed throughout the project rather than a phase conducted at its conclusion (Project Management Institute [PMI], 2021).

4.5 Key Performance Indicators (Donavin)

Introduction

Metrics that indicate whether the California Freeway Contractor Evaluation Framework (CFCEF) project is progressing according to plan are required during execution and monitoring. Key performance indicators (KPIs) are metrics designed to illustrate whether or not a project is producing the desired results. Project boundaries, key deliverables, and the 10-week schedule were defined during Planning. Stakeholder groups and the engagement process were defined during Execution. The following KPIs expand on the groundwork laid during those phases by defining measurable expectations for stakeholder involvement and feedback response, framework quality, deliverable completion, and sponsor acceptance. Defined thresholds provide the project manager with objective information to assess whether project performance is progressing as expected, when performance may be slipping, and if the final CFCEF meets approved project requirements.

Stakeholder Participation

The degree to which stakeholders are involved with framework development can be a key indicator of project success. The stakeholder engagement process defined during Execution identifies Caltrans, contractors and construction firms, the Federal Highway Administration (FHWA), California residents who commute or pay tolls/taxes, and the project team as major stakeholder groups that can provide valuable operational or public insight. Active participation will be tracked by meeting/webinar attendance, survey results, comments posted on the CFCEF Project Page, and written comments received during workshops or review meetings. Tracking participation demonstrates that framework decisions are informed by people who will administer, work under, or be impacted by the procurement framework.

The stakeholder engagement log started during execution will track participation metrics for this KPI. The metric for success is engagement during at least 90% of planned opportunities and documented participation by each major stakeholder group prior to sponsor review. The Stakeholder Engagement Lead will measure the metric at each project milestone and define the KPI status as:

- On Track – when the project is meeting the target
- At Risk – when participation is between 75% and 89%
- Requires Action – when participation drops below 75% or a major stakeholder group has not participated

Approaches to improve stakeholder engagement may be implemented when the KPI is At Risk or Requires Action. Options include increasing visibility on the webpage or newsletter, improving survey distribution options, and providing additional virtual opportunities to engage with the project team. Available options must fall within the approved project scope.

Stakeholder Feedback Disposition

Simply collecting stakeholder feedback is pointless if the project team does not keep a record of how the feedback was considered. For that reason, this KPI measures the percentage of stakeholder feedback that is logged with the engagement plan, assigned to a team member, reviewed, and given a disposition during project execution. Stakeholder recommendations received through surveys, workshops, meetings, email, or the CFCEF Project Page will be logged in the stakeholder engagement log. A team member with relevant responsibilities will be assigned each recommendation. Each stakeholder recommendation will be reviewed and can be accepted, deferred, or rejected based on project scope, regulatory requirements, supporting evidence, and alignment with project objectives. The goal is not to accept every

recommendation, but rather to properly document the disposition of each. Success for this KPI is defined as 100% of stakeholder recommendations having a documented review and decision before final project approval. A completion rate of between 95% and 99% will be considered At Risk and less than 95% Requires Action.

Framework Quality

The primary intended outcome for this project is creation of the California Freeway Contractor Evaluation Framework (CFCEF), so framework quality directly correlates with overall project success. Framework development tasks include peer reviews and sponsor reviews that will evaluate whether the framework being designed will remain objective, realistic, research-supported, and consistent with applicable Caltrans procurement requirements. Areas evaluated during reviews include contractor evaluation factors that are measurable, scoring guidance is clear and consistent, supporting documentation provides an auditable traceability path, and all applicable stakeholder input was considered. The quality target is successful completion of 100% of required review checkpoints with 0 open major quality findings prior to sponsor acceptance. Normal document control processes can be used to make minor revisions. If a change will impact any aspect of approved project scope, schedule, cost, or deliverables it must be submitted through the project's formal change- control process. The KPI will be set to On Track when all review checkpoints have been completed and no major findings are outstanding. At Risk will be used if there are major findings which are pending resolution. Requires Action will be used if there are outstanding findings which will impact a milestone or final acceptance.

Deliverable Completion

In addition to creating an objectively viable and usable CFCEF, project success will be measured by the completion of all deliverables approved during the Planning Phase. These include the current-state assessment document, literature review and industry guidance compilation, Caltrans procurement process analysis, contractor evaluation framework, final report and recommendations, project management plan (inclusive of all required supporting documents), and final presentation. The approved WBS and milestone schedule defined in the planning phase will be used to track planned vs. completed work during weekly project status meetings. The target measure is completion of 100% of approved deliverables by the end of Week 10 of the schedule and successful completion of all required quality reviews for each deliverable. The KPI will be set to On Track if all deliverables are on track to meet their planned milestones. At Risk will be used if a deliverable is forecast to miss a milestone but can be recovered through mitigation strategies without affecting the baseline. Requires Action will be used if a delay will impact the final delivery date in Week 10 or requires a change to the approved project baseline.

Sponsor Acceptance

Acceptance by the CFCEF project sponsor is the final validation that the project team successfully delivered value. Caltrans has been identified as the project sponsor, and as the intended organizational user of the CFCEF, they will provide opportunities to evaluate draft framework components, voice concerns, and recommend revisions during milestone reviews prior to the final presentation. All sponsor feedback will be documented and tracked to resolution. Formal acceptance by the sponsor is the target measure of project success. The deliverable of formal acceptance will require a success metric of completed acceptance with at least 95% stakeholder satisfaction against established review criteria and 0 major concerns

remaining open. The KPI will be set to At Risk while there are any major sponsor concerns outstanding. Once the sponsor has confirmed that the CFCEF meets project objectives and will be used as intended by the project team, KPI will be considered achieved. This metric is specific to acceptance of the contractor selection framework and recommendation; it does not mean that Caltrans has fully implemented, or pilot tested the CFCEF. That determination is beyond the approved scope of this project.

Performance Measurement Summary

Taken collectively, the KPIs outlined above provide metrics to evaluate CFCEF project performance against a balanced set of objectives. Stakeholder participation ensures wide engagement; feedback disposition reinforces accountability; framework quality validates the quality of the primary project output; deliverable completion validates adherence to approved project scope and schedule, and sponsor acceptance validates that the end product will fulfill its intended purpose within the parent organization. Establishing defined targets and thresholds for On Track, At Risk, and Requires Action allows project team members to quickly identify problems before they escalate and correct the process if possible before final project delivery. Leveraging approved project documents including the stakeholder log, WBS, milestones, deliverables, and scope boundaries from the Planning Phase and previously described Execution tasks helps preserve continuity across project phases. According to Proverbs 16: 3 (New International Version), “Commit to the Lord whatever you do, and he will establish your plans.” When describing this project, the Proverb means that team members are accountable for their tasks, reviews are planned, and the project manager will responsibly steward the project towards completion while monitoring the KPIs defined here.

4.6 Conclusion (Ezeoma)

Beyond this project's immediate scope, CFCEF's long-term value lies in its reusability: once adopted, the framework's weighted evaluation criteria and scoring methodology could be applied to future Caltrans highway contracts without being rebuilt each time, giving the agency a durable tool rather than a one-time recommendation. Over multiple contracting cycles, consistently selecting better-qualified contractors could compound into meaningfully lower statewide rework spending, shorter cumulative construction-related disruption for California commuters, and a stronger evidentiary track record that supports further refinement of the model. If successful, CFCEF could also serve as a template other California transportation agencies or infrastructure programs to adapt for their own procurement decisions, extending its impact well past the freeway contracts it was originally designed to evaluate.

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